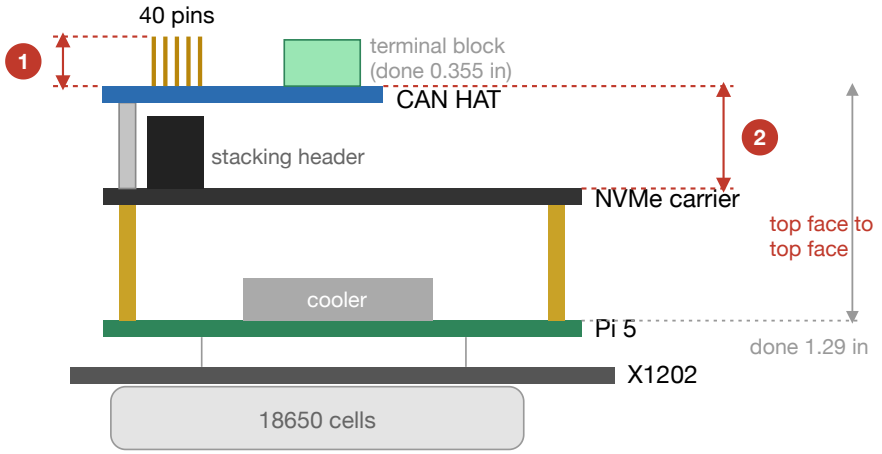
WALL-E Pi 5 shield: measurement sheet

2026-11-14 · Fill in, then send a photo of this sheet. Sketches are not to scale. Numbers in circles match the sketches.

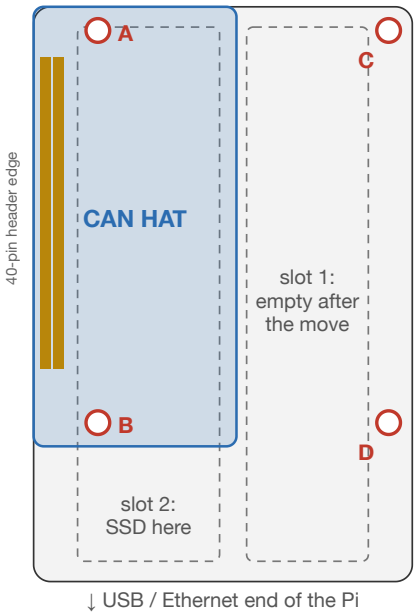
Reading your dial caliper
One full turn of the needle = 1 inch. Each small mark on the dial = 0.01 in.
Whole inches come from the beam. Example: beam past 1, needle on 29 → **1.29 in**.
Write inches. (mm = in × 25.4)

Already done (no need to redo)
Pi top → CAN HAT top: 1.29 in · Terminal block: 0.355 in
Receiver: 1.30 × 0.785 × 0.40 in · Receiver + on Nano 5V
Receiver header: male pins, S + – order

A · The stack: CAN HAT and NVMe carrier



Side view. ① measures the pins sticking up above the CAN HAT. ② is measured beside hole A or B.



Top view, like your photo. A, B: CAN HAT holes. C, D: carrier holes.

1 **Pin height**: CAN HAT top face to the pin tips. Measure at each end of the 40-pin row and in the middle.
Want 0.295 to 0.335 in (7.5 to 8.5 mm)

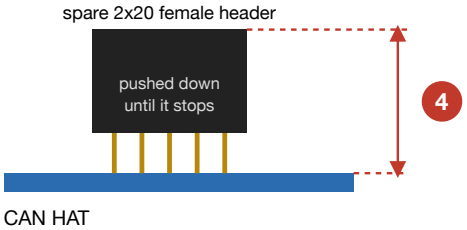
end A _____ middle _____ end B _____ in

2 **Carrier top face to CAN HAT top face**, right beside hole A, then beside hole B.
Sets the long standoffs for C and D (this + 0.433 in)

at A _____ in at B _____ in

3 **What is in each hole today?** Look from the top. Do C and D after the SSD moves to slot 2.

A: ☐ screw head ☐ stud up ☐ empty
B: ☐ screw head ☐ stud up ☐ empty
C: ☐ screw ☐ nut ☐ stud up ☐ empty
D: ☐ screw ☐ nut ☐ stud up ☐ empty

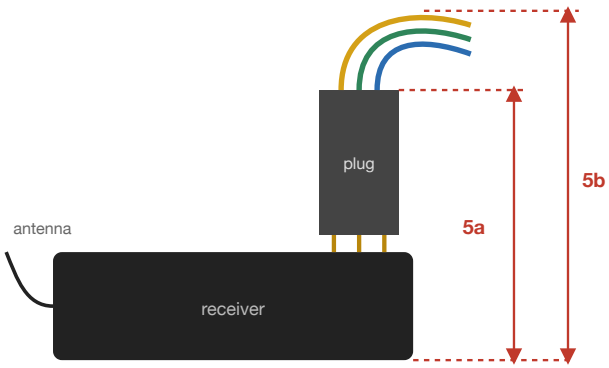


4 **Optional.** Only if you have a spare 2x20 female header (the 8.5 mm tall kind). Push it onto the CAN HAT pins until it stops. CAN HAT top face to the top of the header.
Want 0.433 in (11.0 mm) or less. Skip if you have none.

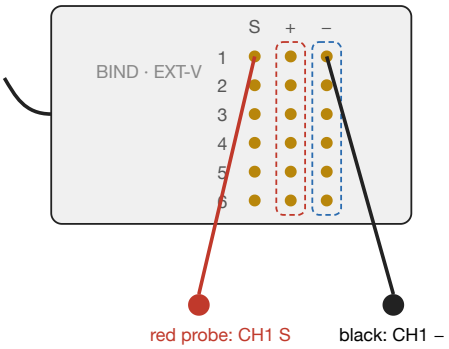
_____ in

B · The receiver (HotRC F-06A)

Side view, plugs on



Top view, plugs off (as labelled on the case)



Dashed boxes: the + column and the – column (check 6)

5 **Height with the plugs on**, from the receiver's bottom face. 5a: to the top of the plug housing. 5b: to the top of the wire bend.

For the case lid. Your harness as it is today.

5a _____ in 5b _____ in

6 **Meter on beep, receiver unplugged from power.** Touch the + pin of row 1, then the + pin of each other row. Then the same for the – pins.

The shield powers the receiver through any one lead, so these must all connect.

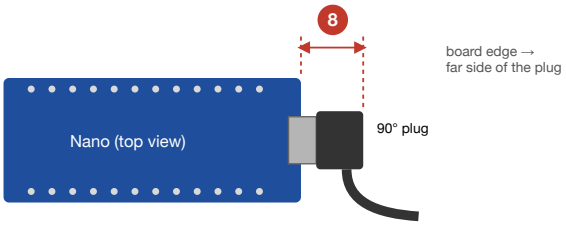
all + beep together? ☐ yes ☐ no
all – beep together? ☐ yes ☐ no

7 **Signal voltage.** Receiver powered (its harness on the Nano, Nano on any USB port), transmitter on, stick centred. Meter on DC volts: red probe on CH1's S pin (or the back of its plug), black on a – pin.

The meter shows the average of the pulses. I work the real level out from it.

_____ V
blue LED on the receiver: ☐ on ☐ off

C · The Nano's USB plug



8 **Your 90° Mini-B plug, plugged into the Nano.** From the Nano's board edge (USB end) to the far side of the plug, or of the cable bend if that sticks out more. _____ in

Nano flat, parts up. Which way does the cable leave the plug? ☐ up ☐ down ☐ left ☐ right ☐ straight out

D · Later, after the board is built

Rover current at first power-up (USB meter) _____ mA

RC range test, receiver on the shield: solid link out to _____ ft

New Nano has the 6-pin ICSP header fitted? (tallest point if so) ☐ yes ☐ no